Relatedness and Structural Dependence

Rheinhessen−Pfalz (DEB3) <−> United States (US) | openalex | log RCA axis; dotted = RCA 1 & median reldens; dashed = log−linear fit

Pearson (log RCA) = 0.06 (p=0.704) Spearman = −0.03 (p=0.876) n = 38

Rheinhessen−Pfalz (DEB3)'s reliance on United States (US) (RCA)

Geothermal Energy
Quantum Technologies
Hydroelectric Power
Pollution Control
Other
Waste Management
Nuclear Energy
High−Performance Computing (HPC)
Environmental Engineering
Carbon Capture & Storage
EVs & Charging Infrastructure
Blockchain Technologies
Battery
Hydrogen Fuel Cells
Water Management
Recycling
Satellite Connectivity
Smart Grid
Green Consumer Goods
Biomass Energy
Green Building
Cybersecurity Technologies
Computer Vision, Language Processing & Object Recognition
Semiconductor Chips
Generative AI
Artificial Intelligence
Software Engineering & System Development
Biofuels
Cloud Computing & Edge Computing
Biomaterials
Drones
Solar Energy
Internet of Things (IoT)
Mobile Networks (5G/6G)
Robotics
Energy Efficiency
Sustainable Agriculture
Heat Pumps

0.2 0.5 1.0 2.0

0 20 40 60 80

Pearson (log RCA) = 0.43 (p=0.007) Spearman = 0.34 (p=0.039) n = 38

United States (US)'s reliance on Rheinhessen−Pfalz (DEB3) (RCA)

Quantum Technologies
Pollution Control
Geothermal Energy
Recycling
Environmental Engineering
High−Performance Computing (HPC)
Waste Management
Generative AI
Battery
Biomaterials
Other
Satellite Connectivity
Computer Vision, Language Processing & Object Recognition
Green Building
Carbon Capture & Storage
Biofuels
Solar Energy
Artificial Intelligence
EVs & Charging Infrastructure
Robotics
Nuclear Energy
Cybersecurity Technologies
Energy Efficiency
Biomass Energy
Drones
Heat Pumps
Sustainable Agriculture

0.2 0.5 1.0 2.0 5.0 10.0

0 20 40 60 80

Relatedness density of Rheinhessen−Pfalz (DEB3)

Pearson (log RCA) = 0.43 (p=0.006) Spearman = 0.35 (p=0.031) n = 38

Rheinhessen−Pfalz (DEB3)'s reliance on United States (US) (RCA)

Geothermal Energy
Quantum Technologies
Hydroelectric Power
Pollution Control
Other
Waste Management
Nuclear Energy
High−Performance Computing (HPC)
Environmental Engineering
Carbon Capture & Storage
EVs & Charging Infrastructure
Hydrogen Fuel Cells
Blockchain Technologies
Recycling
Water Management
Green Consumer Goods
Satellite Connectivity
Smart Grid
Biomass Energy
Computer Vision, Language Processing & Object Recognition
Cybersecurity Technologies
Generative AI
Software Engineering & System Development
Artificial Intelligence
Biofuels
Biomaterials
Cloud Computing & Edge Computing
Drones
Solar Energy
Internet of Things (IoT)
Mobile Networks (5G/6G)
Robotics
Energy Efficiency
Sustainable Agriculture
Heat Pumps

0.2 0.5 1.0 2.0

0 5 10 15 20 25 30

Relatedness density of United States (US)

Pearson (log RCA) = 0.34 (p=0.036) Spearman = 0.35 (p=0.030) n = 38

United States (US)'s reliance on Rheinhessen−Pfalz (DEB3) (RCA)

Quantum Technologies
Pollution Control
Recycling
Geothermal Energy
Hydrogen Fuel Cells
Green Consumer Goods
Environmental Engineering
High−Performance Computing (HPC)
Waste Management
Generative AI
Battery
Biomaterials
Other
Satellite Connectivity
Computer Vision, Language Processing & Object Recognition
Carbon Capture & Storage
Biofuels
Solar Energy
Artificial Intelligence
EVs & Charging Infrastructure
Robotics
Nuclear Energy
Cybersecurity Technologies
Hydroelectric Power
Smart Grid
Drones
Heat Pumps
Sustainable Agriculture

0.2 0.5 1.0 2.0 5.0 10.0

0 5 10 15 20 25 30

Relatedness density of United States (US)